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Updated American Cancer Society lung cancer screening guidelines

The new guidelines offer expanded criteria recommended for lung cancer screening based on age, smoking status, and smoking history

All persons aged 50–80 years who currently or formerly smoked and have at least a 20–pack-year history of smoking (i.e., a pack-year is equal to smoking 1 pack or approximately 20 cigarettes per day for 1 year) should be eligible for annual lung cancer screening with low-dose computed tomography.

These are the new criteria recommended for lung cancer screening in the updated American Cancer Society guidelines for lung cancer screening.[1] Published in November 2023 and based on the most current evidence, the updated guidelines include a number of changes from the previous guidelines published in 2013[2]: age (now 50–80 years vs. 55–74 years before), current or previous smoking status (now all persons in this category vs. those who currently smoke and those who previously smoked who quit within the past 15 years before), and smoking history (now ≥20 pack-years vs. ≥30 pack-years before).

Other changes include which persons should be excluded from screening. The updated guidelines recommend not screening patients for whom comorbid conditions limit life expectancy to <5 years, who are not willing to accept treatment for screen-detected cancer, or whose health conditions may increase harm or interfere with the further evaluation or treatment of lung cancer. Screening exclusions in the 2013 guidelines include having life-limiting comorbid conditions, having implanted metallic devices in the chest or back, and requiring home oxygen.

Both guidelines emphasize the need for shared decision-making (SDM) in helping people to understand the benefits, limitations, and potential harms of low-dose computed tomography and for offering patients smoking-cessation counseling. Both guidelines also recommend offering evidence-based smoking-cessation counseling, with the updated guidelines also recommending offering pharmacotherapy to help those who currently smoke to quit.

Screening all who have smoked regardless of the number of years since quitting

An important difference between the 2023 guidelines and the previous guidelines is the screening eligibility of all people who formerly smoked (aged 50–80 years) who have accumulated at least 20 pack-years regardless of the number of years since quitting. Prior guidelines limited eligibility to those who had quit within 15 years, so patients who had smoked became ineligible for screening 15 years after they had quit.

New evidence indicates that the long-held assumption that lung cancer risk continuously decreases after quitting smoking is erroneous, and the authors of the new guidelines underscore the urgency in correcting the assumption to avert preventable lung cancer deaths in a large group of people for whom screening has not been recommended.

The evidence comes from a modeling study by Landy et al.,[3] who looked at two interrelated issues to quantify the effects of smoking on lung cancer risk. They first looked at the counteracting effects of quit-years and concomitant aging on lung cancer risk in persons with >15 quit-years, and they then used data from prior studies to project the number of preventable lung cancer deaths and life-years gained from screening for a range of quit-year criteria.

They found that the absolute cancer risk increased by 8.7% per year for persons with >15 quit-years of smoking when the counteracting effects of aging and quit-years were considered.

Using data from the National Health Interview Survey (2015–2018) on absolute lung cancer risk and gainable years of life from screening persons aged 50–80 years in the United States who ever smoked, the investigators estimated that 4.9 million more people would be eligible for screening if quit-years were eliminated. Furthermore, they projected that screening all eligible individuals would increase the proportion of preventable lung cancer deaths from 63.7% to 74.2%.

Under an alternative model, the investigators found that the elimination of quit-years from screening recommendations would result in 1.7 million more people being eligible for screening while increasing lung cancer death sensitivity to 74.0%. These estimates were based on augmenting the 2021 US Preventive Services Task Force lung cancer screening criteria with persons who gain the most life-years from screening per the Life Years From Screening CT predictive model.

"Lung cancer screening has been shown to greatly increase the proportion of cancers diagnosed at an early stage, which correlates to greatly improved survival compared to cancers diagnosed at an advanced stage," says Rebecca Landy, PhD, a research fellow at the National Cancer Institute and lead author of the modeling study. "Individuals who have quit smoking many years earlier may still be at a relatively high risk of lung cancer and should have an SDM session with a health care professional to decide whether the benefits of lung cancer screening are likely to outweigh its potential harms."

Increasing screening uptake

One key takeaway from the new guidelines is the need for better uptake of screening for eligible people. Lung cancer screening rates have been and continue to be very low. A 2022 report by the American Lung Association showed that only 5.8% of eligible persons underwent lung cancer screening, with screening rates ranging from 1% in California to 16.3% (the highest rate) in Massachusetts.[4]

The panel of investigators emphasized that, among the various reasons for this poor uptake, the primary reason that determines whether people are screened is the advice that they receive, or do not receive, from their health care provider. This points to the importance of health care providers being aware of which patients are eligible for lung cancer screening and then engaging in SDM with their patients with respect to the benefits, limitations, and potential harms of cancer screening. Among the core elements of SDM is educating patients on what the screening test is and how it is performed, how often they should be screened, who is eligible, and what the benefits of screening are as well as any health status that may preclude screening benefits, any harms/limitations of screening, and the importance of smoking cessation.

The authors point out, however, that strategies are needed to help providers and practices to identify patients eligible for screening. Although electronic health records (EHRs) currently do not help to identify people who are eligible on the basis of their smoking history, adapting EHRs to do this and improving their functionality would make them an effective tool. The authors suggest that these changes to EHRs may follow the completion by the National Center for Quality Assurance of new lung cancer screening measures in the Healthcare Effectiveness Data and Information Set, which is projected for the end of 2024.[5]

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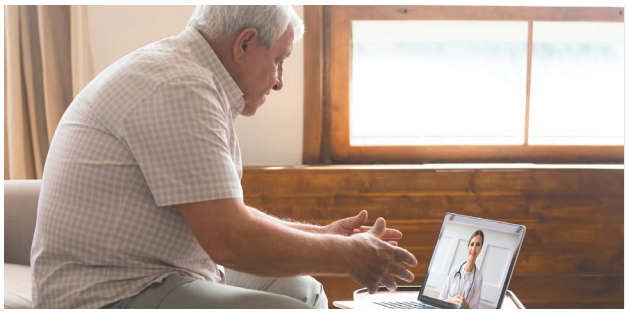
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Continuous expansion of telehealth in cancer care

Telehealth visits for cancer care grew continuously from 2020 to 2021 without any evidence of duplicative care at a multisite, multiregional cancer practice with a diverse patient population according to a study from investigators at the Mayo Clinic in Rochester, Minnesota, that looked at telehealth visits for cancer care beyond the initial expansion of telehealth during the COVID-19 pandemic.[1]

"Telehealth adoption has continued to grow beyond the immediate pandemic period, and patients with cancer are as satisfied with telehealth visits as they are with facility-based care," says the lead author of the study, Joshua C. Pritchett, MD, a hematology/oncology fellow at the Mayo Clinic. No increases in per-patient or per-provider visit volumes were seen with the growth of telehealth adoption. "This implies that telehealth visits can be integrated into cancer care delivery without increasing overall visits or duplicative care," he says.



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Conducting the study over 8-week periods from July to August 2019, from July to August 2020, and from July to August 2021, the investigators found that the rate of telehealth use for outpatient cancer visits increased from <0.01% in 2019 (n = 32,537) to 14% in 2021 (n = 35,820) among the Mayo Clinic's medical oncology and hematology practices, which are set in three geographically diverse tertiary referral campuses.

The study found that the most significant factors linked with the increased use of telehealth visits for cancer were being

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